

## BURERAM, Thailand

WMO: 484370

Lat: 15.2274N

Lon: 103.2440E

Elev: 604

StdP: 14.38

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | 58.1       | 61.0 | 49.3                          | 53.1 | 64.1 | 51.7 | 58.2 | 65.6 | 15.6                  | 78.8 | 13.7 | 80.0 | 2.0                   | N/A  | 0.297 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 18.4                   | 100.6           | 77.8 | 98.2 | 77.8 | 96.1 | 77.9 | 82.1                | 90.7 | 81.6 | 90.2 | 81.0 | 89.6 | 3.6                  | N/A  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 80.1                            | 159.7 | 86.1 | 79.4 | 156.2 | 85.4 | 78.9 | 153.5 | 85.0 | 46.5          | 90.8 | 45.8 | 90.5 | 45.3 | 89.9 | 85.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 11.9          | 9.8  | 8.2 | 52.8                       | 103.0 | 2.1                | 2.8 | 51.3                                               | 105.0 | 50.1       | 106.6 | 48.9       | 108.1 | 47.4       | 110.1 |    |    |
| (5)               |      |     | 50.3                       | 83.7  | 2.5                | 0.9 | 48.5                                               | 84.4  | 47.0       | 84.9  | 45.6       | 85.4  | 43.8       | 86.1  |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 81.4   | 75.0  | 79.3  | 84.1  | 86.4  | 85.3  | 84.6  | 83.2  | 82.5  | 81.7  | 80.7  | 78.7  | 75.0 |      |
|                                                                                  | DBStd       | 5.17   | 4.77  | 5.57  | 5.04  | 4.23  | 2.97  | 2.20  | 2.51  | 1.97  | 2.12  | 2.06  | 3.46  | 4.81 |      |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |      |
|                                                                                  | HDD65       | 4      | 2     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1    |      |
|                                                                                  | CDD50       | 11457  | 776   | 822   | 1058  | 1092  | 1093  | 1037  | 1030  | 1009  | 952   | 952   | 862   | 774  |      |
|                                                                                  | CDD65       | 5985   | 313   | 402   | 593   | 642   | 628   | 587   | 565   | 544   | 502   | 487   | 412   | 310  |      |
|                                                                                  | CDH74       | 62797  | 2601  | 4342  | 7350  | 8182  | 7399  | 6657  | 6038  | 5525  | 4691  | 4254  | 3363  | 2395 |      |
|                                                                                  | CDH80       | 27079  | 934   | 2050  | 3922  | 4463  | 3510  | 2848  | 2393  | 1979  | 1515  | 1416  | 1222  | 827  |      |
| Wind                                                                             | WSAvg       | 2.1    | 2.1   | 1.8   | 2.1   | 2.2   | 1.7   | 2.4   | 2.9   | 2.7   | 1.8   | 1.9   | 1.7   | 2.3  |      |
| Precipitation                                                                    | PrecAvg     | 54.1   | 0.4   | 0.6   | 1.8   | 3.3   | 6.8   | 7.5   | 8.5   | 9.5   | 10.3  | 4.5   | 0.9   | 0.1  |      |
|                                                                                  | PrecMax     | 68.0   | 2.5   | 2.4   | 5.3   | 8.1   | 10.2  | 12.6  | 14.2  | 15.1  | 18.1  | 9.3   | 4.0   | 0.7  |      |
|                                                                                  | PrecMin     | 45.5   | 0.0   | 0.0   | 0.1   | 1.1   | 3.3   | 3.3   | 3.1   | 3.5   | 4.6   | 0.1   | 0.0   | 0.0  |      |
|                                                                                  | PrecStd     | 5.4    | 0.6   | 0.6   | 1.4   | 1.6   | 1.9   | 2.2   | 3.1   | 2.7   | 3.2   | 2.3   | 1.0   | 0.2  |      |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 92.6  | 97.6  | 102.9 | 105.4 | 100.3 | 98.1  | 97.2  | 93.1  | 92.3  | 92.0  | 93.1  | 92.3 |      |
|                                                                                  |             | MCWB   | 74.6  | 75.8  | 75.7  | 77.8  | 79.6  | 80.3  | 78.9  | 80.4  | 80.9  | 79.5  | 78.5  | 75.0 |      |
|                                                                                  | 2%          | DB     | 90.3  | 95.5  | 99.8  | 101.6 | 97.8  | 95.2  | 94.0  | 91.5  | 90.7  | 89.9  | 90.6  | 90.1 |      |
|                                                                                  |             | MCWB   | 73.7  | 74.9  | 75.7  | 78.2  | 79.6  | 80.2  | 79.6  | 80.2  | 80.6  | 78.8  | 77.6  | 74.7 |      |
|                                                                                  | 5%          | DB     | 87.9  | 93.5  | 97.4  | 98.8  | 95.5  | 92.9  | 91.7  | 90.0  | 89.2  | 88.5  | 88.9  | 87.6 |      |
|                                                                                  |             | MCWB   | 72.5  | 74.6  | 76.1  | 78.2  | 79.8  | 80.1  | 79.5  | 79.8  | 80.2  | 78.3  | 76.9  | 74.1 |      |
|                                                                                  | 10%         | DB     | 85.5  | 90.9  | 94.8  | 96.0  | 93.1  | 90.9  | 90.9  | 89.7  | 88.5  | 87.5  | 87.1  | 87.0 | 85.0 |
|                                                                                  |             | MCWB   | 71.8  | 74.4  | 76.2  | 78.0  | 79.8  | 80.0  | 79.3  | 79.5  | 79.7  | 77.9  | 76.1  | 72.8 |      |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 78.1  | 78.9  | 79.8  | 81.8  | 82.7  | 83.1  | 81.9  | 81.9  | 82.6  | 81.8  | 80.8  | 77.9 |      |
|                                                                                  |             | MCDB   | 88.3  | 92.0  | 91.9  | 94.7  | 92.3  | 91.8  | 90.5  | 89.7  | 89.3  | 89.1  | 88.6  | 88.4 |      |
|                                                                                  | 2%          | WB     | 76.5  | 77.6  | 78.7  | 80.7  | 81.8  | 81.9  | 81.1  | 81.1  | 81.6  | 80.6  | 79.4  | 76.3 |      |
|                                                                                  |             | MCDB   | 85.7  | 90.5  | 91.4  | 94.4  | 91.4  | 90.8  | 89.6  | 88.6  | 88.2  | 87.3  | 87.9  | 86.5 |      |
|                                                                                  | 5%          | WB     | 75.0  | 76.5  | 78.0  | 79.8  | 81.0  | 81.2  | 80.4  | 80.6  | 80.8  | 79.7  | 78.2  | 75.2 |      |
|                                                                                  |             | MCDB   | 83.4  | 89.1  | 90.9  | 93.1  | 90.9  | 89.9  | 88.7  | 87.9  | 87.4  | 86.0  | 86.2  | 85.1 |      |
|                                                                                  | 10%         | WB     | 73.6  | 75.5  | 77.2  | 79.0  | 80.4  | 80.4  | 79.7  | 80.0  | 80.1  | 78.8  | 77.2  | 73.9 |      |
|                                                                                  |             | MCDB   | 82.3  | 87.5  | 90.4  | 91.8  | 90.2  | 88.9  | 87.5  | 86.6  | 86.0  | 84.7  | 83.9  | 82.6 |      |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 20.8  | 21.5  | 19.9  | 18.4  | 15.7  | 14.1  | 13.2  | 12.4  | 11.9  | 13.3  | 16.5  | 19.3 |      |
|                                                                                  |             | MCDBR  | 21.8  | 22.8  | 22.4  | 21.8  | 19.0  | 16.8  | 16.2  | 14.4  | 14.0  | 15.1  | 18.0  | 20.2 |      |
|                                                                                  |             | MCWBR  | 7.4   | 6.7   | 5.1   | 4.7   | 4.6   | 4.9   | 5.0   | 4.9   | 5.4   | 5.6   | 6.8   | 7.2  |      |
|                                                                                  | 5% WB       | MCDBR  | 18.7  | 20.2  | 19.6  | 19.5  | 16.3  | 14.8  | 14.3  | 13.4  | 13.4  | 13.7  | 16.2  | 19.0 |      |
|                                                                                  |             | MCWBR  | 7.3   | 6.3   | 5.7   | 5.6   | 5.3   | 5.6   | 5.4   | 5.3   | 5.8   | 6.0   | 6.6   | 8.2  |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.488  | 0.605 | 0.659 | 0.602 | 0.503 | 0.465 | 0.457 | 0.460 | 0.498 | 0.525 | 0.477 | 0.467 |      |      |
|                                                                                  | taud        | 2.052  | 1.769 | 1.692 | 1.861 | 2.176 | 2.292 | 2.316 | 2.295 | 2.177 | 2.071 | 2.182 | 2.149 |      |      |
|                                                                                  | Ebn at Noon | 254    | 230   | 222   | 235   | 256   | 263   | 266   | 267   | 258   | 248   | 256   | 257   |      |      |
|                                                                                  | Edh at Noon | 52     | 71    | 79    | 67    | 48    | 42    | 41    | 43    | 48    | 52    | 45    | 46    |      |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1582   | 1663  | 1697  | 1823  | 1800  | 1699  | 1613  | 1556  | 1495  | 1578  | 1571  | 1535  |      |      |
|                                                                                  | RadStd      | 88     | 72    | 69    | 75    | 98    | 69    | 105   | 99    | 98    | 107   | 68    | 43    |      |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (17 neighbors) | +0.47 | N/A     | N/A    | N/A     | N/A   | +0.45 | +0.40       | N/A   | N/A   | +185  |

Nomenclature: See separate page